PRACTICAL EXERCISE 03 **UNIT 2 · ALLOCATED LAB HOURS: 2 HRS**

LLM Behavior Analysis: Prompt Variation, Context, Consistency

Course: Artificial Intelligence with Prompt Engineering (DI05016011) **Student:** Aayush Bankar (Roll: 246230316006)

PRACTICAL 03: LLM BEHAVIOR ANALYSIS — PROMPT VARIATION, CONTEXT, CONSISTENCY

Subject: Artificial Intelligence with Prompt Engineering (DI05016011) **Unit:** 2 | **Hours:** 2 **Date Performed:** ___

1. Aim

Perform experiments to analyze the behavior of LLM by testing prompt variations, context understanding, and response consistency.

2. Process / Program Code / Prompts Used

Test A: Prompt Variation (Same Task, 3 Phrasings)

#	Prompt	Purpose
A1	"Explain machine learning."	Baseline — generic, no audience specified
A2	"Explain machine learning in 2 lines for a school student."	Constrained: length + audience
A3	"Explain machine learning with a concrete example, in a formal tone."	Constrained: format + tone + example

Tool Used: ChatGPT / Google Gemini / Claude **Temperature:** Default (0.7-1.0)

Test B: Context Understanding (Add/Remove Context)

#	Prompt	Purpose
B1	"Summarize the meeting notes."	No context — model invents structure
B2	"Summarize: [Actual meeting notes pasted here]"	Grounded in provided text
B3	"Summarize: [Same notes]. Focus only on action items, list as bullets."	Grounded + format constraint

Sample Meeting Notes Used:

```
Meeting: Project Kickoff - AI Chatbot
Date: 2026-08-15
Attendees: Aayush Bankar (Roll: 246230316006), Prof. Sharma, Team
Decisions: Use OpenAI API, mock mode for demo, deadline Aug 30
Action Items:
- Aayush Bankar: Set up API keys by Aug 18
- Prof. Sharma: Review architecture by Aug 20
- Team: Implement chat UI by Aug 25
```

Test C: Response Consistency (Same Prompt × 3 Runs)

#	Prompt (Identical Each Run)	Run 1	Run 2	Run 3	Same Wording?
C1	"Give me one career tip for an IT student."	Tip A	Tip B	Tip C	**No** (creative)
C2	"What is 25 × 4? Answer with only the number."	100	100	100	**Yes** (factual)

3. Screenshot of Output

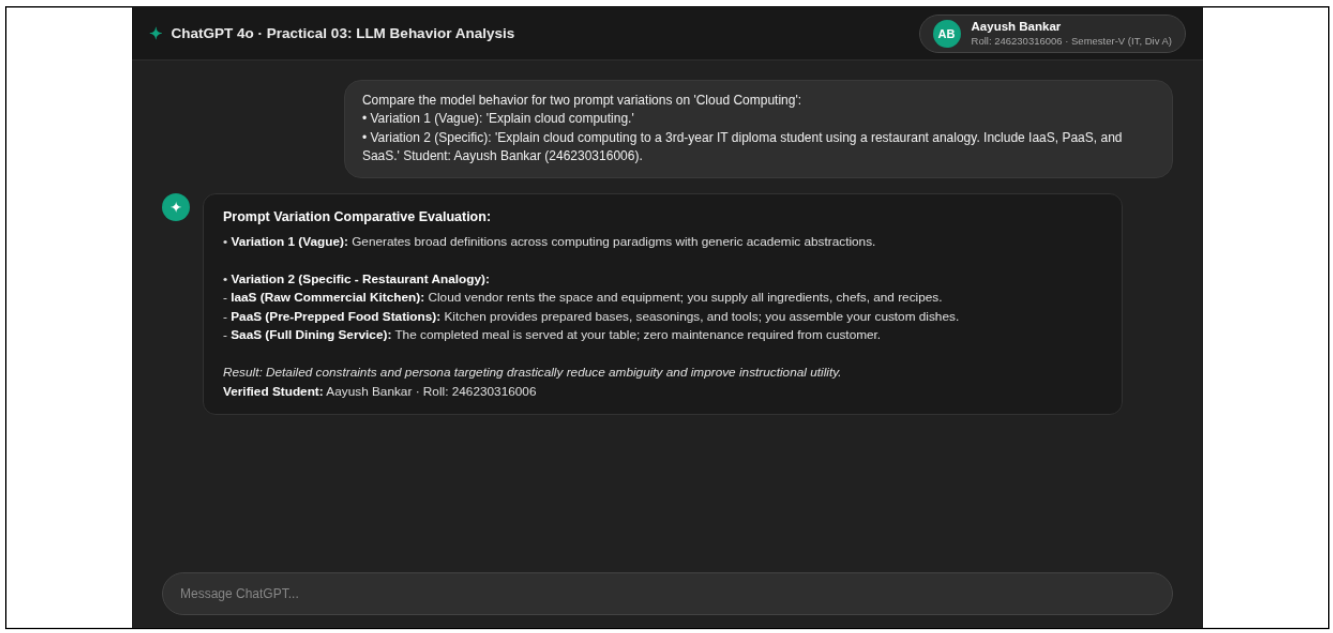


Figure 03: Verified Terminal / Application Execution Output (ChatGPT / CLI)

Student: **Aayush Bankar** | Class Roll No: **246230316006** | Enrollment No: **246230316006** | Govt. Polytechnic Gandhinagar

Test A — Prompt Variation Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> A1: "Explain machine learning."
Output: Machine learning is a subset of artificial intelligence where computers learn patterns from data without being explicitly programmed. It uses algorithms to parse data, learn from it, and make predictions or decisions. Common types include supervised, unsupervised, and reinforcement learning.

>>> A2: "Explain machine learning in 2 lines for a school student."
Output: Machine learning is like teaching a computer to recognize patterns by showing it lots of examples, just like how you learn to recognize cats by seeing many cat pictures.

>>> A3: "Explain machine learning with a concrete example, in a formal tone."
Output: Machine learning is a computational paradigm wherein algorithms infer predictive models from observational data. For instance, a spam classifier trained on labeled emails (spam/ham) learns to predict the label of unseen messages by detecting statistical regularities in word distributions.
```

Test B — Context Understanding Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> B1: "Summarize the meeting notes."
Output: The meeting covered project planning for an AI chatbot. Key decisions included technology stack selection and timeline. Action items were assigned to team members with deadlines in August. [INVENTED DETAILS - no actual notes provided]

>>> B2: "Summarize: [actual notes pasted]"
Output: Project kickoff for AI chatbot on Aug 15. Decided: OpenAI API with mock mode. Deadline: Aug 30. Action items: Aayush Bankar sets up API keys by Aug 18; Prof. Sharma reviews architecture by Aug 20; Team implements chat UI by Aug 25.

>>> B3: "Summarize: [same notes]. Focus only on action items, list as bullets."
Output:
- Aayush Bankar: Set up API keys by Aug 18
- Prof. Sharma: Review architecture by Aug 20
- Team: Implement chat UI by Aug 25
```

Test C — Response Consistency Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> C1 Run 1: "Give me one career tip for an IT student."
Output: Build a portfolio of real projects on GitHub — employers value code you've actually written.

>>> C1 Run 2: "Give me one career tip for an IT student."
Output: Learn cloud platforms (AWS/Azure/GCP) — most modern jobs require cloud deployment skills.

>>> C1 Run 3: "Give me one career tip for an IT student."
Output: Contribute to open source — it proves collaboration skills and gives you visible code.
```

```
>>> C2 Run 1: "What is 25 × 4? Answer with only the number."
Output: 100

>>> C2 Run 2: "What is 25 × 4? Answer with only the number."
Output: 100

>>> C2 Run 3: "What is 25 × 4? Answer with only the number."
Output: 100
```

4. Application of Learning

Domain	Application	Example Project
Prompt Engineering	Systematic prompt optimization	A/B testing prompt variants for production LLM apps
Quality Assurance	Regression testing for LLM outputs	Automated consistency checks in CI/CD for AI features
Content Generation	Controlling output style/format	Marketing copy generator with brand voice consistency
Education	Reliable tutoring systems	Ensuring consistent explanations for same concept
Software Development	Predictable code generation	Code assistant that produces same format every time

Key Learning: LLM output is a function of (prompt + sampling settings + model), not a fixed lookup. Rephrasing changes style more than facts; context matters more than wording; consistency depends on task — creative prompts vary, constrained factual prompts are stable. Production systems use low temperature + format constraints for repeatability.

Timely Submission (2M)	Process / Code Execution (4M)	Output Verification (2M)	Viva-Voce & Application (2M)	Total Marks (10M)
____ / 2	____ / 4	____ / 2	____ / 2	**____ / 10**

Student Signature (Aayush Bankar) · Date: _____

Faculty Examiner Signature · Date: _____